

On Stablecoin Price Processes and Arbitrage^{*}

Ingolf Gunnar Anton Pernice

Weizenbaum Institute, Berlin 10623, Germany

Abstract. This study applies the Caginalp & Balenovic (1999) model for asset flow dynamics to fully collateralized stablecoins. The analysis provides novel insights on how trend-reversion and reactions to peg deviations work together to keep stablecoin prices close to the price they are targeting. A fixed-effects panel regression indicates that the model’s abstraction of trading motivations indeed fits stablecoin price processes well. The results convey first indication that theoretic stablecoin models might benefit from modeling price dynamics to switch between two market regimes: one for day-to-day price formation and limited arbitrage activity; and one for extraordinary market situations.

Keywords: stablecoins · arbitrage · price formation

1 Introduction

Stablecoins are being increasingly adopted as bridge to trade traditional cryptocurrencies [4, 9, 40, 48], marketed as medium-of-exchange for decentralized finance and smart contracts [41, 42] and have recently been approved by US regulators as payment method for federally chartered banks [1]. Increasing practical relevance fosters the demand for understanding economic properties of such instruments. The majority of projects simply tokenize the asset their tokens are stabilized against (e.g. the USD, EUR or gold) or store a third asset in the respective amount [45]. Assuming that traders trust governance, collateral and technology, and their trust is justified, there is little room for disagreement on the token’s fundamental values. If there were any structured deviation, efficient-market theorists might argue that market participants would seize the resulting opportunity of arbitrage, closing the gap [26].¹ Stablecoin prices would then merely reflect the value of the collateral and random noise. If, however, price adjustments are restricted by incomplete arbitrage, interesting patterns might emerge that reflect the trader’s decisions [10, 22]. This paper explores and quantifies such dynamics.

^{*} I thank Gunduz Caginalp for his invaluable input and enlightening conversations. I also thank Martin Florian and Anna Almosova for their constructive feedback.

¹ In a strict sense, arbitrage opportunities can be defined as “investment strategy that guarantees a positive payoff in some contingency with no possibility of a negative payoff and with no net investment” [25, p.57]. In this paper the term is used in a wider sense, describing the trader’s perceptions.

My empiric approach is based on a recent theoretical approach from asset pricing literature proposed first in [18] and refined by [14] which opens up asset pricing to dynamic systems modeling known from thermodynamic physics. In the following, the latter will be called the “Caginalp & Balenovic (1999) asset flow” (*CBAF*) approach. In a nutshell, the authors model price determination by abstracting trading decisions as flows from asset-to-cash and vice versa. As such, traders are abstracted as being driven by price trends and the deviation of the asset’s market price from its fundamental value. Exploring theoretic approaches to the instability of price processes, a first application to the field of cryptocurrencies has been attempted by [12]. Applying the *CBAF* model to stablecoin arbitrage promises to offer a convincing model for the interplay of trend following and peg deviations. In contrast to most models for cryptocurrency pricing, the above model requires few assumptions.² Exploring early trading data for 11 fully collateralized stablecoins, this study evaluates the appropriateness of the intuitive trader abstraction adopted by the *CBAF* approach and offers insights into stablecoin price dynamics in general. My empirical setup couples variables approximating trend-following with a measure for peg deviations in a dynamic coin-fixed-effects (coin-FE) panel data regression. The study engages in rigorous robustness checking by testing the models for the influence of seasonal dummies, interaction terms for the direction of trends, sign of peg deviations and different parameters in data preprocessing. I find a striking difference between results being based on data with—and without outlier treatment even for merely truncating price changes exceeding 5 standard deviations. For data including extreme price changes, the *CBAF* model approximates price formation well. Deviations from the peg and trend following are strong determinants of coming price changes. For outlier-free data, however, the effects seem to blend in with other price-determinants (the stablecoin’s token supply and Bitcoin’s price volatility). This result indicates that apart from very few occasions arbitrage activity is weak. This poses the questions, whether costs of arbitrage might be prohibitive high except for extreme but rare market situations. It might be considered to model price formation by switching between two regimes: a first one offering profits for large trades towards the peg, and a second one being characterized by limited levels of arbitrage activity.

The paper is organized as follows: Section 2 outlines related work, Section 3 introduces the theoretic backgrounds, Section 4 discusses data and econometric approach. Section 5 describes the results and Section 6 concludes the study.

2 Related Work

Studies investigating the asset class and its manifestations include [45], [19], [43], [23] and [11]. The user perception in the adoption of stablecoins has been studied in [34], while [6] analyzed the suitability of stablecoins as save-haven investment. The relation between cryptocurrency prices and stablecoin trading

² Compared with, for example, currency competition models [2, 27, 28, 47], game-theoretic approaches [13] and consumer demand models [3, 5, 8].

has been focused on by [9], [4], [48] and [29]. Adopting a risk-oriented approach, [35], [36] and [37] suggest theoretic models to study stability and resilience of stablecoins. While the latter focuses on extreme events, [40] not only offers a theoretic model for day-to-day arbitrage but also provides, to my knowledge, the first extensive empirical analysis of the drivers of stablecoin prices. This paper differs in perspective and econometric approach: Instead of peg deviations, this paper analyzes price processes and tests the *CBAF* model for applicability. While [40] relies on a rich database, however, just for *Tether*, this paper adopts a dynamic coin-FE panel data regression on market data for 11 stablecoins.

3 Stablecoin arbitrage and the CBAF model

Stablecoin arbitrage uses primary and secondary markets [40]. On the primary market, coins are created and redeemed against collateral with the issuer. On the secondary market, stablecoins are traded against fiat and cryptocurrencies. If market prices deviate from the peg, arbitrage traders might decide to either trade against peg deviations directly on the secondary market or involve the stablecoin issuer. In the first case, arbitrage traders simply trade towards the peg. They would buy when prices are below—and sell when prices are above the peg. In the second case, abstracting from technical details,³ arbitrage traders would purchase coins from the markets and redeem them with the issuer when prices are below the peg. When prices are above the peg, traders would first create coins with the issuer and subsequently sell them on secondary markets.

The *CBAF* approach as presented in [14] models traders as switching from cash to asset or vice versa with a certain probability k . Variable k is modeled to include motivations based on past price changes ζ_1 and the market discount relative to the asset's fundamental value ζ_2 . These components are modeled using the trader's memory with respect to price trends and deviations from fundamental values (c_1 and c_2) and their focus on these two respective components (q_1 and q_2). Core component of ζ_1 is the relative price trend $f(\tau) = \frac{1}{P(\tau)} \frac{dP(\tau)}{d\tau}$ over period τ on time scale τ_0 . ζ_2 is constructed around deviations $g(\tau) = \frac{P_a(\tau) - P(\tau)}{P(\tau)}$ of market price $P(\tau)$ from fundamental value P_a . Thus, the two components can be expressed as

$$\zeta_1(t) = q_1 c_1 \int_{-\infty}^t e^{\frac{-(t-\tau)}{c_1}} \cdot f(\tau) d\tau \quad (1) \quad \zeta_2(t) = q_2 c_2 \int_{-\infty}^t e^{\frac{-(t-\tau)}{c_2}} \cdot g(\tau) d\tau. \quad (2)$$

Limiting k to values between 0 and 1, k is constructed as $k = \frac{1}{2} + \frac{\tanh(\zeta_1 + \zeta_2)}{2}$. For ζ_1 and ζ_2 equal to zero, the probability of flows from cash to asset and vice versa are thus equally likely. Demand D and supply S of the asset are modeled using k and the fraction already invested into the asset B . Their relation can be expressed as $D = k(1 - B)$ and $S = (1 - k)B$ and thus $\frac{D}{S} = \frac{k}{1-k} \frac{1-B}{B}$. Prices P

³ Models for more complex stablecoins can be found e.g. in [35] and [36].

are assumed to change logarithmic with excess demand, leading to

$$\frac{1}{P} \frac{dP}{dt} = \delta \cdot \log\left(\frac{D}{S} - 1\right) = \delta \cdot \log\left(\frac{k}{(1-k)} \frac{1-B}{B}\right) \quad (3)$$

with δ representing an amplitude that scales with time. For a deeper discussion of the model and its applications see [14], [46], [14], [17], [16] and [12]. While this paper will not be offering estimates for the individual parameters, the study gathers evidence supporting the basic intuition of the model: Under the efficient market hypothesis, fluctuations of $P(t)$ around the peg, assumed to equal P_a , would merely be random.⁴ Obviously, in this case, neither of the two components ought to be reflected in market data of stablecoins. If, however, evidence of ζ_1 or ζ_2 is present, it is of considerable interest to understand how the two components are jointly driving stablecoin prices.

4 Data & Econometric Approach

The following paragraphs will discuss data preprocessing and the economic approach for testing the applicability of the *CBAF* model.

Data was gathered from www.coingecko.com. Similar to www.coinmarketcap.de the data provider currently crawls 382 cryptocurrency exchanges but offers hourly data over a well-documented API for free.⁵ The full dataset comprises 19 cryptocurrencies. The sample includes projects that are listed at www.coingecko.com, promise stability of their exchange rates in their whitepapers and collateralize their tokens to at least 100%. The study considers designs that use the asset-pegged-to as collateral but also includes tokens that use a third asset, often a crypto-asset, in a quantity reflecting at least full collateralization. Some of the 19 cryptocurrencies are quite young and immature. Shallow markets with low volumes and few trades per measurement period might in part be driven by market microstructure effects which could lead to biased regressions. To reduce noise, this study thus excludes months for that the stablecoin shows a market capitalization of under USD 10m or daily trading volumes of under USD 1m. Coins have been dropped completely if the remaining dataset included less than $24 \cdot 31$ hourly observations. This restricted the dataset to 11 stablecoins with 767 to 16970 trading hours leading to 101,243 observations in total. Table 6 of Appendix 7.3 shows that the results hold as well if market capitalization thresholds of USD 100m and USD 5m—and thresholds for daily trading volumes of USD 50m and USD 50k are chosen. To understand the effect of extreme values on the estimated models, price changes diverging over 5 standard deviations (SD) from the mean have been truncated. Table 2 in Appendix 7.2 gives an overview over cutoff points and the relative and absolute number

⁴ This study assumes that traders rightfully trust in the peg as a correct estimate of the tokens fundamental value. This fails when doubts about the stablecoins collateral or security arise.

⁵ As of 2020-08-17.

of classified outliers. The threshold was chosen deliberately high to correct only the most extreme values. Such might have a disproportionate effect on ordinary-least-squares (OLS) regression results [7]. To make the estimated coefficients comparable, all variables apart from the dummies have been standardized based on Z-scores $X^{stand} = \frac{X - \mu(X)}{\sigma(X)}$, with mean $\mu(\cdot)$ and standard deviation $\sigma(\cdot)$.

To verify the applicability of the *CBAF* model, a linear panel regression in conjunction with squared and cubed trend variables to model nonlinear relationships has been applied to exchange-traded funds [17] and stocks [15]. Now this framework is applied to stablecoins by adjusting it slightly to the characteristics of the dataset and adding a variety of robustness checks. Traditional linear modeling might fail to pick up the complex relations in price formation. In contrast to linear models, polynomials allow for very strong negative (positive) past returns to induce positive (negative) bounce-back effects. Mixing up relations between strong and weak price changes, simple auto-regressive price regressions abstaining from the above step might miss potential information on prices. As a remedy, following [15] and [17], this study includes variables to the power into a fixed-effects panel data approach.⁶ To capture price trends, in accordance to [15], this paper uses a simple weighted aggregation of past price changes. The prior might be expressed using prices P and smoothing factor s over a look-back window d as

$$T_t = \frac{1}{\sum_{k=1}^d e^{sk}} \sum_{k=1}^d \left(\frac{P_{t-k+1} - P_{t-k}}{P_{t-k}} \cdot e^{sk} \right). \quad (4)$$

The smoothing term ensures that older observations of variable changes are entering the sum with lower weights ($s = -0.25$ and $d = 10$). Trends in prices have been shown to explain variation in future returns for cryptocurrencies in [20], [32] and [30]. To calculate the distance D of market price P from the peg P_a , simply $D_t = P_t - P_{a,t}$ is formed. As control variables the token supply and Bitcoin's price-volatility⁷ are used. Regressions are formed denominating cryptocurrency price changes as ΔP , the trend as T and peg deviations as D . Moreover ΔS and ΔV^{BTC} are the first differences of token supply and Bitcoin's price-volatility respectively. Dummy variables include seasonal ones (Z_t^{hour} , Z_t^{day} , Z_t^{month}) and others that account for the sign of the peg deviation ($Z_t^{D>0}$) and the direction of the trend ($Z_t^{T>0}$). The unobserved coin effect is denominated as a_i while b_t gives the unobserved time effect and the remaining residual errors are given as u_{it} . A parsimonious base-line regression can thus be based on unobserved-effects equation

$$\begin{aligned} \Delta P_{i,t+1} = & \beta_2 T_{i,t} + \beta_3 T_{i,t}^2 + \beta_4 T_{i,t}^3 + \beta_5 D_{i,t} + \beta_3 D_{i,t}^2 + \beta_4 D_{i,t}^3 + \\ & \beta_5 \Delta S_{i,t} + \beta_6 \Delta V^{\text{BTC}} + a_i + b_t + u_{it}, \end{aligned} \quad (5)$$

⁶ Robustness against multicollinearity among the regressors is ensured by checking the respective Variance Inflation Factors (VIF) (compare Appendix 7.3 Table 3).

⁷ Controlling for the volatility of the second and third largest traditional cryptocurrencies by trading volume (*Ether* and *Ripple*) or for the *Ethereum Gas* price turned out to yield insignificant coefficients.

while the full general regression is based on

$$\begin{aligned} \Delta P_{i,t+1} = & \beta_2 T_{i,t} + \beta_3 T_{i,t}^2 + \beta_4 T_{i,t}^3 + \beta_5 D_{i,t} + \beta_6 D_{i,t}^2 + \beta_7 D_{i,t}^3 + \\ & \beta_8 \Delta S_{i,t} + \beta_9 \Delta V_{i,t}^{\text{BTC}} + Z_t^{\text{hour}} + Z_t^{\text{day}} + Z_t^{\text{month}} + \beta_{10} T_{i,t} \cdot Z_t^{T>0} + \\ & \beta_{11} T_{i,t} \cdot Z_t^{D>0} + \beta_{12} D_{i,t} \cdot Z_t^{D>0} + \beta_{13} D_{i,t} \cdot Z_t^{T>0} + a_i + b_t + u_{it}, \end{aligned} \quad (6)$$

where t is the time- and i the coin index.

While two-way fixed-effects regression are applied to eliminate time and entity effects in the original framework [15, 17], recent research indicates that this approach might lead to mostly uninterpretable coefficients [31, 38] and biased inferences in most general applications [33]. This study thus settles on eliminating a_i by coin-fixed-effects but models common time effects by including seasonal dummies and control variables. Treatment of the Nickel bias and other issues related to panel data regressions with long time series dimensions (e.g. heteroskedasticity, non-stationarity and serial correlation) are treated in line with state-of-the-art approaches. For more information refer to Appendix 7.1.

5 Results

Table 1 supplies estimates of the coefficients given in Equation 4. As suggested by the applied asset pricing theory, not only current deviations from the peg but also price trends show significant relations with the coming hour's price change. Also, the adopted nonlinear regression framework has proven useful. Most of the variables that are raised to the second and third power show high significance and our models explain roughly 20 % of the variance in the one-hour-ahead price changes. However, coefficient estimates vary with the way outliers are treated.

For the model based on data for which no outlier treatment has been implemented (compare Column 1 Table 1), the polynomial of significant coefficients for the estimated price change is $\widehat{\Delta P}_{t+1} = -0.160D_t + 0.085D_t^2 - 0.417D_t^3$. This polynomial is displayed in Figure 1, which relates peg deviations and trends to the coming hours price change. Even relative small peg deviations are associated with moderate price changes. Large peg deviations, however, precede extreme price changes forcefully driving prices back to the peg. The coefficients related to price trends display nonlinearity as well and form $\widehat{\Delta P}_{t+1} = -0.266D_t + 0.178D_t^3$. Weak price trends seem to be reverted, while stronger ones lead to trend following. In other words, series of consecutive strong price changes of the same direction are prolonged. This might be a result of consecutive hours of price jumps after large deviations. In comparison to price trends, but also all other tested variables, the joint influence of the coefficients for peg deviations seems to be the largest by far. Their influence is roughly symmetric for positive and negative price deviations. For the model based on data for which price changes in excess of 5 SD have been truncated (Column 2 of Table 1), the joint estimated effect of price trends and peg deviations on coming price changes is displayed in Figure 2. For peg deviations, the polynomial of significant coefficients is $\widehat{\Delta P}_{t+1} = -0.259D_t + 0.046D_t^2 + 0.056D_t^3$ with roots roughly at -2.6 , 0 and

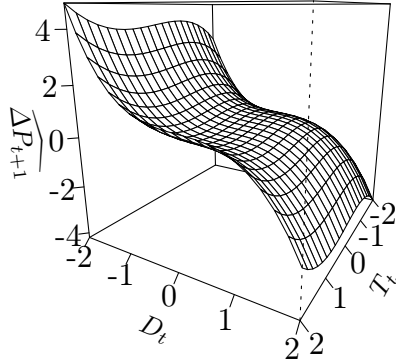
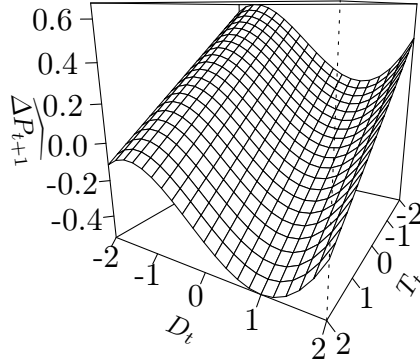
Table 1. Coin-FE regression based on Equation 4.

	<i>Dependent variable:</i>	
	ΔP_{t+1}	
	(1)	(2)
D_t	-0.160*** (0.008)	-0.259*** (0.008)
D_t^2	0.085*** (0.027)	0.046*** (0.006)
D_t^3	-0.417*** (0.046)	0.056** (0.023)
T_t	-0.266*** (0.009)	-0.184*** (0.005)
T_t^2	-0.009 (0.023)	-0.005 (0.008)
T_t^3	0.178*** (0.053)	-0.011 (0.037)
ΔS_t	0.208*** (0.077)	0.394*** (0.144)
ΔV_t^{BTC}	-0.070 (0.050)	-0.308*** (0.089)
Observations	101,243	101,243
R ²	0.200	0.198
Adjusted R ²	0.200	0.198
F Statistic	3,162.209***	3,123.468***
(df = 8; 101224)		

Note:

*p<0.1; **p<0.05; ***p<0.01

Variable denominations are given in Section 4 under Equation 4. Column (1) for untreated- and column (2) for outlier free data. Standard errors given in brackets.

**Fig. 1.** Coefficient interplay for untreated data**Fig. 2.** Coefficient interplay for outlier-free data

1.8. The effect of smaller peg deviations is in line with expectations: The relation of peg deviations and the coming price change is positive for $-2.6 \leq D \leq 0$ and negative for $0 \leq D \leq 1.8$ and would thus lead to mean-reverting prices. For positive deviations $D > 1.8$ and negative deviations $D < -2.6$, however, signs turn and peg deviations are associated with price changes leading away from the peg. The reason for this seemingly counter-intuitive result might be that the majority (95.37%) of observed price changes lie within the 2 SD displayed in Figures 1 and 2. Rather than avoiding rare but large errors from outliers, the coefficients seem to be optimized to fit smaller fluctuations around the peg. Not only the results for peg deviations—but also price trends differ. The latter, estimated based on outlier free data, show a linear, negative relation to coming price changes. In other words, series of price changes of equal sign revert soon. The size of the coefficients for trend and peg deviations are relatively low, though. For instance, a peg deviation of 2 SD in negative direction is associated merely with an increase in prices of 0.1 SD. A negative price trend of 1 SD is associated with an increase of prices of around 0.2 SD. Arbitrage is estimated to have a very limited effect in the outlier-free dataset. With coefficients of 0.394 and -0.308 respectively, the change in token supply and the change in the Bitcoin’s price volatility seem to be equally important price determinants, at least. This contrast is surprising, given that they are caused by the truncation of merely 363 of 101243 observations. Table 5 of Appendix 7.3 gives coefficients for the same model but complemented by dummy representations accounting for hours, weekdays, months as well as interaction terms for the sign of the peg deviation and the direction of the price trend. The results differ only negligibly.

6 Conclusion

The stark contrast of the importance of peg deviations in the fitted models with—and without outlier treatment is striking. For data including extreme price changes, peg deviations and trend dynamics seem to approximate price formation well. For stablecoins, the intuitive *CBAF* approach to model trader behavior might thus be seen as a good approximation for the determinants of price changes. For outlier-free data, however, the effects seem to blend in with other price-determinants. Which is the right approach to take, when describing stablecoin price dynamics? The above results pose the question of whether stablecoin markets might not best be modeled as switching between two different regimes: one characterized by limited arbitrage, and one setting in when markets require correction and promise profits for large trades towards the peg. It might be considered to adjust the *CBAF* model to stablecoin arbitrage by including a certain threshold of tolerance to peg deviations. Future research might analyze whether the above results are a consequence of the costs for arbitrage.

7 Appendix

7.1 Robustness

The dataset applied in this study combines 11 timeseries of differing lengths and might thus be described as an unbalanced timeseries panel. While a large T dimension is generally beneficial, simple panel data approaches might be misspecified. A first issue is serial correlation. In most financial time series prior realizations affect coming ones. Including lagged data might thus be useful to capture serial correlation in the data - this is usually referred to as dynamic panel modeling. Instead of including lagged data explicitly, in this study, the trend variable is carrying auto-regressive information.⁸ Using simple fixed-effects models jointly with lagged variables, however, induces the so-called Nickell bias as the lagged variable causes endogeneity in the regressors [44]. As argued by [21, p.163], including fixed-effects into dynamic specifications of panel data regressions, even for simple OLS estimates, can mitigate the issue to some degree. Their coefficients, however, are still seriously biased for small T . In our case, including coin-fixed-effects and considering that T is very large, Nickell's bias should be negligible.⁹ There are other issues known from time-series analysis, though. [49] warned about relying on the above for inference for non-stationary data (which might lead to spurious regression results) and suggested to check the error term for heteroskedasticity, serial correlation and nonnormality. To counter this problem, this study ensures stationarity using the Levin-Lin-Chu unit root test [39]. As the test does not reject the presence of a unit root for token supply and volatility, we take first differences of these variables.

As discussed earlier, we apply coin-FE panel regressions based on simple OLS-estimation. As a consequence, several assumptions are to be ensured. Residuals ought to display a mean of zero and be free of heteroscedasticity, cross-sectional, and serial correlation. Breusch-Pagan Lagrange Multiplier tests and Pesaran cross-sectional dependence tests are used to test for cross-sectional dependence in the residuals. Additionally, Student's t-tests have been applied to check the residuals for a mean of zero. Breusch-Godfrey/Wooldridge tests have been applied to test for serial correlation. Breusch-Pagan tests are used for detecting heteroskedasticity. While a deviation from zero for the residuals is strongly rejected, unfortunately, the remaining tests reveal heteroscedasticity, serial, and also cross-sectional correlation. In other words, residuals are showing variance clusters and are depending on their own- and even lags across coins. As a consequence, the simple OLS estimator is biased. To still draw robust inferences from the estimated model, spacial correlation consistent (SCC) estimators introduced in [24] are used. The approach adapts Newey-West estimators to the panel setting and leads to robust standard errors even in the presence of heteroscedasticity and cross-sectional and serial correlation.

⁸ As discussed earlier, the trend-variable is constructed as a weighted average of a certain number of past lags.

⁹ In fact, following [49], the bias for the fixed-effects estimator approaches zero with rate $1/T$.

7.2 Preparatory Empirics and Descriptives

Table 1. Descriptives.

<i>Projects</i>	<i>Obs.</i>	σ	<i>Vol.</i>	<i>Supply</i>	<i>Peg Dev.</i>		
					<i>Min.</i>	<i>Avg.</i>	<i>Max.</i>
Tether	16970	0.24	107.51	2067.48	-2.48	0.03	2.55
True-USD	16919	0.42	32.54	205.80	-4.18	0.05	4.29
Paxos Standard	16907	0.36	63.81	137.03	-3.55	0.03	3.61
USD-Coin	16320	0.37	205.23	309.42	-3.48	0.08	3.90
USDK	10358	0.56	30.23	28.56	-4.05	-0.28	4.76
Maker Dai	8195	1.10	2.78	41.78	-4.08	0.91	12.07
HUSD	5226	0.21	56.76	135.18	-2.33	-0.02	2.28
Binance USD	4466	0.16	208.15	176.59	-1.87	-0.01	1.84
Gemini Dollar	2869	0.62	3.76	15.91	-4.05	0.13	3.06
NUSD	2268	1.69	5.67	38.03	-2.76	1.33	15.37
MUSD	767	0.40	3.40	52.48	-2.73	-0.09	1.39

Data truncated at 5 standard deviations. Boundaries for market capitalization and 24h trading volumes of USD 10m and USD 1m. *Obs.* is the number of observations, σ the standard deviation. *Vol.* is the 24h trading volume and *Supply* the token supply— both in millions. *Peg Dev.* is the peg deviations scaled by factor 100.

Table 2. Outliers.

Cutoff (in Std. Dev.)	Outliers as Defined by Cutoff	
	N	%
1	18839	18.6
2	4686	4.63
3	1563	1.54
4	683	0.67
5	363	0.36
6	192	0.19
7	104	0.1
8	52	0.05
9	32	0.03
10	26	0.03

Fraction of data classified as outliers. Data pre-processing as in Table 1 of Appendix Section 7.2.

Table 3. Variance inflation factors.

	S_t	D_t	D_t^2	D_t^3	V_t^{BTC}	T_t	T_t^2	T_t^3
Binance USD	2.00	1.52	1.48	1.99	2.01	1.51	1.61	1.95
Maker Dai	3.33	2.31	6.12	4.77	3.63	1.40	1.28	1.52
Gemini Dollar	1.32	2.13	1.28	2.15	1.45	2.07	1.31	2.22
HUSD	4.26	1.56	2.11	2.03	4.32	1.52	1.91	1.98
MUSD	1.12	2.14	2.37	3.61	1.14	2.24	1.65	2.62
NUSD	1.51	6.96	31.12	17.00	1.30	1.97	1.23	1.82
Paxos Standard	2.52	1.98	1.97	2.75	2.55	1.65	1.10	1.54
Tether	1.89	1.56	1.48	1.76	1.91	1.53	1.24	1.54
True-USD	1.14	1.94	2.19	2.17	1.26	1.34	1.07	1.23
USD-Coin	1.67	2.14	2.48	2.25	1.99	1.28	1.23	1.47
USDK	1.38	3.71	1.62	3.03	1.44	1.37	1.11	1.38

This table gives the variance inflation factors. Data preprocessing as in Table 1 of Appendix Section 7.2. As all factors are well below the rule of thumb value of 10, it is concluded that the risk of multicollinearity is negligible. Variable denominations are given in Section 4 under Equation 4.

Table 4. Unit root tests.

	P_t	T_t	D_t	S_t	V_t^{BTC}
χ^2	1717.89	723.38	1717.89	7.09	2.27
df	22.00	22.00	22.00	22.00	22.00
p	0.00	0.00	0.00	1.00	1.00

This table gives the results of the unit root test for the data truncated at 5 standard deviations. Data preprocessing as in Table 1 of Appendix Section 7.2. Variable denominations are given in Section 4 under Equation 4. Coefficient's standard errors given in brackets.

7.3 Robustness Checks

Table 5. Coin-FE regression.

<i>Dependent variable:</i>		
	ΔP_{t+1}	
	(1)	(2)
D_t	-0.111*** (0.010)	-0.230*** (0.014)
D_t^2	0.089*** (0.027)	0.047*** (0.007)
D_t^3	-0.472*** (0.047)	0.032 (0.025)
T_t	-0.248*** (0.012)	-0.180*** (0.009)
T_t^2	-0.010 (0.023)	-0.006 (0.009)
T_t^3	0.162*** (0.054)	-0.015 (0.038)
ΔS_t	0.202*** (0.074)	0.392*** (0.142)
ΔV_t^{BTC}	-0.063 (0.051)	-0.304*** (0.089)
Z_t^{hour}	-0.0003 (0.0003)	-0.0003 (0.001)
Z_t^{day}	-0.0002 (0.0003)	-0.001 (0.001)
Z_t^{month}	-0.003*** (0.0003)	-0.006*** (0.001)
$D \cdot Z_t^{D>0}$	-0.018*** (0.002)	-0.006 (0.009)
$D \cdot Z_t^{T>0}$	-0.003** (0.001)	-0.011 (0.007)
$T \cdot Z_t^{D>0}$	0.003*** (0.001)	-0.005 (0.008)
$T \cdot Z_t^{T>0}$	0.0001 (0.001)	0.009 (0.007)
Observations	101,243	101,243
R ²	0.208	0.200
Adjusted R ²	0.208	0.199
F Statistic	1,769.824***	1,681.830***
(df = 15; 101217)		

Note: *p<0.1; **p<0.05; ***p<0.01

This table gives the results of a coin-FE regression for the full model specified in Equation 4. Variable denominations are given in Section 4 under Equation 4. Column (1) for untreated- and column (2) for outlier free data. Coefficient's standard errors given in brackets.

Table 6. Coin-FE regression for full model based on dataset with strict or weak restriction rules.

	<i>Dependent variable:</i>			
	(1)	(2)	(3)	(4)
D_t	-0.331*** (0.025)	-0.366*** (0.019)	-0.095*** (0.008)	-0.252*** (0.011)
D_t^2	0.451*** (0.075)	0.122*** (0.015)	-0.031* (0.017)	0.001 (0.005)
D_t^3	-0.291*** (0.092)	0.048* (0.025)	-0.110*** (0.043)	0.109*** (0.011)
T_t	-0.206*** (0.014)	-0.176*** (0.013)	-0.184*** (0.008)	-0.134*** (0.007)
T_t^2	-0.118*** (0.042)	-0.029*** (0.011)	-0.001 (0.015)	0.005 (0.005)
T_t^3	0.014 (0.091)	0.029 (0.051)	0.059 (0.041)	0.013 (0.019)
ΔS_t	0.273*** (0.066)	0.775*** (0.180)	0.145*** (0.022)	0.019*** (0.002)
ΔV_t^{BTC}	-0.155** (0.069)	-0.670*** (0.149)	-0.027 (0.044)	-0.207** (0.085)
Z_t^{hour}	0.00005 (0.0003)	0.0004 (0.001)	0.0004 (0.0002)	0.001 (0.0005)
Z_t^{day}	-0.001*** (0.0003)	-0.002** (0.001)	-0.0004* (0.0002)	-0.001* (0.0005)
Z_t^{month}	-0.001*** (0.0004)	0.0003 (0.001)	-0.001*** (0.0003)	-0.003*** (0.001)
$D \cdot Z_t^{D>0}$	0.006*** (0.002)	0.039*** (0.013)	-0.010*** (0.001)	0.035*** (0.006)
$D \cdot Z_t^{I>0}$	-0.006*** (0.002)	-0.032*** (0.009)	-0.001 (0.001)	0.016*** (0.005)
$T \cdot Z_t^{D>0}$	-0.006*** (0.001)	-0.038*** (0.013)	-0.00002 (0.001)	-0.037*** (0.006)
$T \cdot Z_t^{I>0}$	0.003*** (0.001)	0.029*** (0.009)	-0.002*** (0.001)	-0.017*** (0.005)
Adjusted R ²	0.229	0.248	0.140	0.169
F Statistic	1,261.372***	1,401.268***	1,546.137***	1,934.048***

Results for the estimation of Equation 4 with stricter (Column 1 and 2) or weaker data restrictions (Column 3 and 4). The dataset is restricted with market capitalization thresholds of 100m and 5m USD —and thresholds for trading volumes of 50m and 50k USD. Variable denominations are given in Section 4 under Equation 4. Column (1) and (3) for untreated- and column (2) and (4) for outlier free data. Coefficient's standard errors given in brackets.

References

1. Office of the Comptroller of the Currency federally chartered banks and thrifts may participate in independent node verification networks and use stablecoins for payment activities. <https://www.occ.gov/news-issuances/news-releases/2021/nr-occ-2021-2.html>, accessed: 2021-01-07
2. Almosova, A.: A note on cryptocurrencies and currency competition (1792)
3. Almosova, A.: A monetary model of blockchain (2018)
4. Ante, L., Fiedler, I., Strehle, E.: The influence of stablecoin issuances on cryptocurrency markets. *Finance Research Letters* p. 101867 (2020)
5. Athey, S., Parashkevov, I., Sarukkai, V., Xia, J.: Bitcoin pricing, adoption, and usage: Theory and evidence (2016)
6. Baumöhl, E., Vyrost, T.: Stablecoins as a crypto safe haven? not all of them! (2020)
7. Belsley, D.A., Kuh, E., Welsch, R.E.: Regression diagnostics: Identifying influential data and sources of collinearity, vol. 571. John Wiley & Sons (2005)
8. Biais, B., Bisiere, C., Bouvard, M., Casamatta, C., Menkveld, A.J.: Equilibrium bitcoin pricing. Available at SSRN (2018)
9. Bianchi, D., Iacopini, M., Rossini, L.: Stablecoins and cryptocurrency returns: Evidence from large bayesian vars. Available at SSRN (2020)
10. Britten-Jones, M., Neuberger, A.: Arbitrage pricing with incomplete markets. *Applied Mathematical Finance* **3**(4), 347–363 (1996)
11. Bullmann, D., Klemm, J., Pinna, A.: In search for stability in crypto-assets: are stablecoins the solution? ECB Occasional Paper (230) (2019)
12. Caginalp, C.: A dynamical systems approach to cryptocurrency stability. arXiv preprint arXiv:1805.03143 (2018)
13. Caginalp, C., Caginalp, G.: Establishing cryptocurrency equilibria through game theory. *Mathematics (AIMS)*, Forthcoming (2019)
14. Caginalp, G., Balenovich, D.: Asset flow and momentum: deterministic and stochastic equations. *Philosophical Transactions of the Royal Society of London. Series A: Mathematical, Physical and Engineering Sciences* **357**(1758), 2119–2133 (1999)
15. Caginalp, G., Desantis, M.: Stock price dynamics: nonlinear trend, volume, volatility, resistance and money supply. *Quantitative Finance* **11**(6), 849–861 (2011)
16. Caginalp, G., DeSantis, M.: Nonlinear price dynamics of s&p 100 stocks. *Physica A: Statistical Mechanics and its Applications* p. 122067 (2019)
17. Caginalp, G., DeSantis, M., Sayrak, A.: The nonlinear price dynamics of us equity etfs. *Journal of econometrics* **183**(2), 193–201 (2014)
18. Caginalp, G., Ermentrout, G.: A kinetic thermodynamics approach to the psychology of fluctuations in financial markets. *Applied Mathematics Letters* **3**(4), 17–19 (1990)
19. Clark, J., Demirag, D., Moosavi, S.: Sok: Demystifying stablecoins. Available at SSRN 3466371 (2019)
20. Corbet, S., Eraslan, V., Lucey, B., Sensoy, A.: The effectiveness of technical trading rules in cryptocurrency markets. *Finance Research Letters* **31**, 32–37 (2019)
21. Croissant, Y., Millo, G., et al.: Panel data econometrics with R. Wiley Online Library (2019)
22. Delbaen, F., Schachermayer, W.: The mathematics of arbitrage. Springer Science & Business Media (2006)
23. Dell’Erba, M.: Stablecoins in cryptoeconomics from initial coin offerings to central bank digital currencies. *NYUJ Legis. & Pub. Pol’y* **22**, 1 (2019)

24. Driscoll, J.C., Kraay, A.C.: Consistent covariance matrix estimation with spatially dependent panel data. *Review of economics and statistics* **80**(4), 549–560 (1998)
25. Dybvig, P.H., Ross, S.A.: Arbitrage. In: *Finance*, pp. 57–71. Springer (1989)
26. Fama, E.F.: Random walks in stock market prices. *Financial analysts journal* **51**(1), 75–80 (1995)
27. Fernández-Villaverde, J., Sanches, D.: Can currency competition work? *Journal of Monetary Economics* **106**, 1–15 (2019)
28. Garratt, R., Wallace, N.: Bitcoin 1, bitcoin 2,.....: An experiment in privately issued outside monies. *Economic Inquiry* **56**(3), 1887–1897 (2018)
29. Griffin, J.M., Shams, A.: Is bitcoin really untethered? *The Journal of Finance* **75**(4), 1913–1964 (2020)
30. Grobys, K., Ahmed, S., Sapkota, N.: Technical trading rules in the cryptocurrency market. *Finance Research Letters* **32**, 101396 (2020)
31. Hill, T.D., Davis, A.P., Roos, J.M., French, M.T.: Limitations of fixed-effects models for panel data. *Sociological Perspectives* **63**(3), 357–369 (2020)
32. Hudson, R., Urquhart, A.: Technical trading and cryptocurrencies. *Annals of Operations Research* pp. 1–30 (2019)
33. Imai, K., Kim, I.S.: On the use of two-way fixed effects regression models for causal inference with panel data. Unpublished paper: Harvard University (2019)
34. Kimmerl, J.: Understanding users’ perception on the adoption of stablecoins-the libra case. In: *PACIS*. p. 187 (2020)
35. Klages-Mundt, A., Harz, D., Gudgeon, L., Liu, J.Y., Minca, A.: Stablecoins 2.0: Economic foundations and risk-based models. In: *Proceedings of the 2nd ACM Conference on Advances in Financial Technologies*. pp. 59–79 (2020)
36. Klages-Mundt, A., Minca, A.: (in) stability for the blockchain: Deleveraging spirals and stablecoin attacks. *arXiv preprint arXiv:1906.02152* (2019)
37. Klages-Mundt, A., Minca, A.: While stability lasts: A stochastic model of stablecoins. *arXiv preprint arXiv:2004.01304* (2020)
38. Kropko, J., Kubinec, R.: Why the two-way fixed effects model is difficult to interpret, and what to do about it. Available at SSRN 3062619 (2018)
39. Levin, A., Lin, C.F., Chu, C.S.J.: Unit root tests in panel data: asymptotic and finite-sample properties. *Journal of econometrics* **108**(1), 1–24 (2002)
40. Lyons, R.K., Viswanath-Natraj, G.: What keeps stablecoins stable? Tech. rep., National Bureau of Economic Research (2020)
41. Misc.: Centre whitepaper. <https://www.centre.io/pdfs/centre-whitepaper.pdf>, visited on 2018-11-30
42. Misc.: Stably whitepaper. <https://s3.ca-central-1.amazonaws.com/stably-public-documents/whitepapers/Stably+Whitepaper+v6.pdf> (2018), visited on 2018-07-16
43. Moin, A., Sekniqi, K., Sirer, E.G.: Sok: A classification framework for stablecoin designs. In: *Financial Cryptography* (2020)
44. Nickell, S.: Biases in dynamic models with fixed effects. *Econometrica: Journal of the econometric society* pp. 1417–1426 (1981)
45. Pernice, I.G., Henningsen, S., Proskalovich, R., Florian, M., Elendner, H., Scheuermann, B.: Monetary stabilization in cryptocurrencies—design approaches and open questions. In: *2019 Crypto Valley Conference on Blockchain Technology (CVCBT)*. pp. 47–59. IEEE (2019)
46. Porter, D.P., Smith, V.L.: Stock market bubbles in the laboratory. *Applied mathematical finance* **1**(2), 111–128 (1994)
47. Schilling, L., Uhlig, H.: Some simple bitcoin economics. Tech. rep., National Bureau of Economic Research (2018)

48. Wang, G.J., Ma, X.y., Wu, H.y.: Are stablecoins truly diversifiers, hedges, or safe havens against traditional cryptocurrencies as their name suggests? *Research in International Business and Finance* p. 101225 (2020)
49. Wooldridge, J.M.: *Introductory econometrics: A modern approach*. Nelson Education (2016)